Modelling Fresnel Diffraction from an Aperture Using Numerical Integration

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INTRODUCTION

Augustin-Jean Fresnel was a French physicist who developed one of the most important concepts in waves and optics, called Fresnel diffraction. Fresnel diffraction describes the way light behaves when it goes through obstacles or slits/apertures. This might sound similar to Fraunhofer diffraction but they differ slightly. Fraunhofer diffraction, looks at the diffraction of light at large distances whereas Fresnel diffraction is the exact opposite. It approximates the diffraction of light close to the source/object. The discoveries of Fresnel diffraction has many applications from microscope to computer hard disks, DVDs, etc.[1, 2]

This report aims to describe a mathematical model to compute and approximate Fresnel diffractions of a light beam through apertures of various shapes and sizes, The approximation is computed by numerical integration method. The first part of the model describes diffraction through a one dimensional slit and the second part describes diffractions through square, rectangle and circle apertures of various sizes.

THEORY

The general formula for Fresnel diffraction, which is an expression for the electric field of the diffracted light onto a screen parallel to the aperture is given by

$$E(x,y,z) = \frac{e^{ikz}}{i\lambda z}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} E(x',y') \times \exp\left\{\frac{ik}{2z}[(x-x')^2+(y-y')^2]\right\}dx'dy' \tag{1}$$

where $k = 2\pi/\lambda$ is the wavenumber, λ is the wavelength, z is the screen distance and x, y - x', y' are coordinates on the screen and the aperture respectively.

The form of the Fresnel diffraction equation used in this model is given by

$$E(x,y,z) = \frac{kE_0}{2\pi z}\iint_{\text{Aperture}} E(x',y') \times \exp\left\{\frac{ik}{2z}[(x-x')^2+(y-y')^2]\right\}dx'dy' \tag{2}$$

where E_0 is the electric field of light at the aperture in V/m. The intensity of the diffracted electric field is given by

$$I(x,y,z) = \epsilon_0 c E(x,y,z)E^*(x,y,z) \tag{3}$$

where ϵ_0 is the permittivity of free space, c is the speed of light, E is the electric field of diffracted light with E^* as its complex conjugate.
The diffraction of light is dependent on its wavelength and the aperture width, as the aperture width (slit width) gets closer to the λ of the incident light, a greater diffraction will be observed.
As z gets greater the intensity of the diffracted light goes down. This is due a phenomenon called wavefront spreading or similarly inverse square law. As light waves pass through the aperture they will start to diverge outwards and spread out, this is famously called diffraction. The total energy that the incident light has does not change in this model therefore as the diffracted pattern covers more space on the screen, less energy per unit area will be shone onto the screen. Inverse square law similarly states the same thing. Intensity in general is inversely proportional to the square of the distance from the energy source. This can also be seen in the *eqs.1-2*. [3]

METHOD

Throughout this model quadrature method in SciPy library was used to numerically integrate the Fresnel integrals. When integrating the Fresnel equation in both one and two dimensions, the equation was split into its real and imaginary parts.

Part A: Fresnel Diffraction in 1D

For diffraction in one dimension, since the aperture and the screen have no height the integral and the part of the function that include y in *eq. 2* were not considered. The integration was computed using '`quad()`' with starting and ending points of the aperture, $-x'$ and x' taken as lower and higher limits of integration respectively.
The effects the aperture width, $(x'_1 - x'_2)$ and the screen distance, z on the diffraction pattern done by varying the two parameters. The number of equally spaced points were set to 500 for $z < 0.001m$ and 200 for anything higher. The screen coordinates were adjusted accordingly. (Check appendix for the full equation used for part A *eq.4*.)

Part B: Fresnel Diffraction in 2D

In this part the heights of the aperture and the screen were taken into consideration, and the `dblquad()` function was used to compute the double integral with $x'(y')$ and $x'(y')$ being the inner integral, and y_1' and y_2' being the outer integral limits, lower and higher respectively.

In part B three different 2D shapes (square, rectangle and circle) were utilised to investigate their effects on the diffraction pattern, with the function for $x'(y')$ and $x'(y')$ changing for each shape. The sizes of the square and the rectangle apertures were set by user input for the aperture width, with the rectangle having an aspect ratio of 1:3. The circle apertures size was changed by various inputs for the radius of the circle. The number of points for each side was set to 100 since anything higher caused very long run times.

All the constant used in the model are shown in *table 1* which can be found in the Appendix.

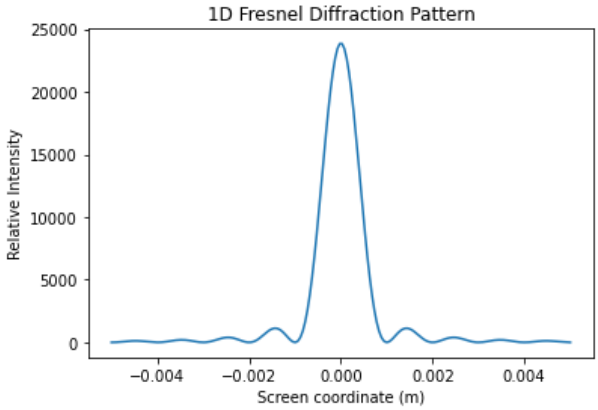


FIG. 2. Fresnel diffraction pattern in one dimension with an aperture width of 2x$10^{-5}m$, and screen distance of $0.02m$.

RESULTS AND ANALYSIS

Looking at *figs.1-2* it is easily seen that the intensity increased as the screen got closer to the aperture. This trend continues and is also existent in *figs.3-4* and in *figs. 5-7* the intensity starts to decrease again as the screen starts to get further away from the aperture.

Part A:

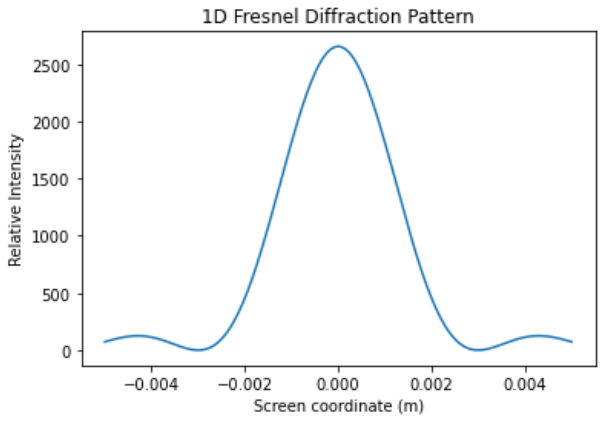


FIG. 1. Fresnel diffraction pattern in one dimension with an aperture width of 2x$10^{-5}m$, and screen distance of $0.06m$.

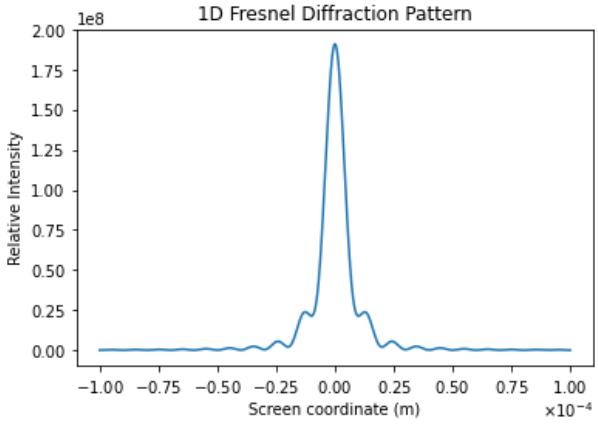


FIG. 3. Fresnel diffraction pattern in one dimension with an aperture width of 2x$10^{-5}m$, and screen distance of 2x$10^{-4}m$.

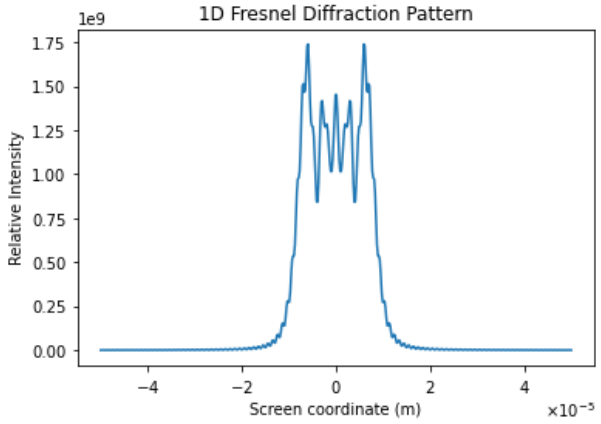


FIG. 4. Fresnel diffraction pattern in one dimension with an aperture width of 2x10^{-5}m, and screen distance of 2x10^{-5}m.

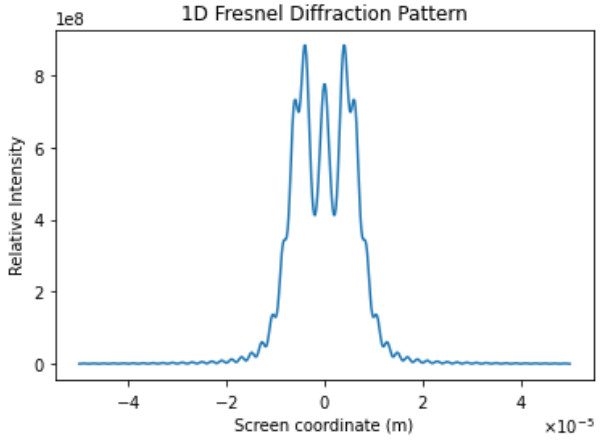


FIG. 5. Fresnel diffraction pattern in one dimension with an aperture width of 2x10^{-5}m, and screen distance of 4x10^{-5}m.

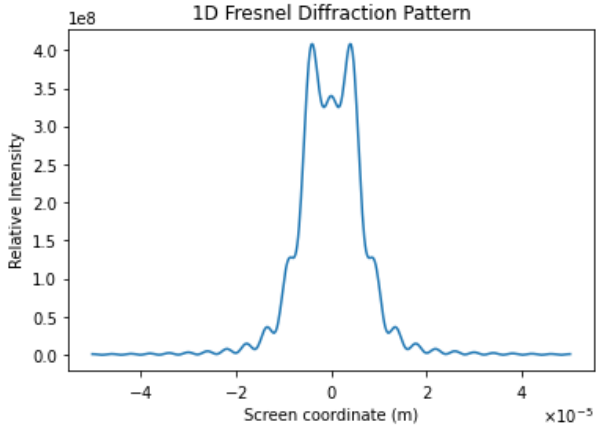


FIG. 6. Fresnel diffraction pattern in one dimension with an aperture width of 2x10^{-5}m, and screen distance of 8x10^{-5}m.

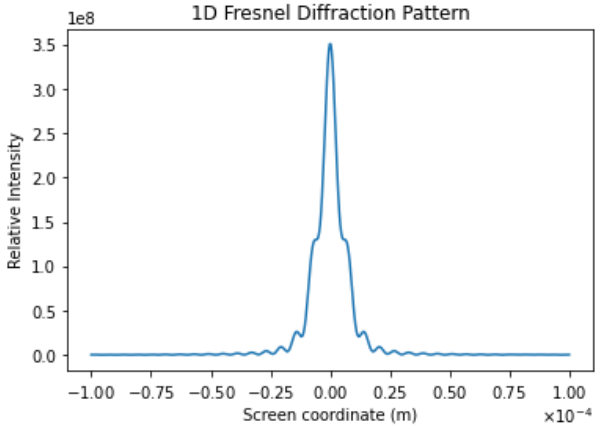


FIG. 7. Fresnel diffraction pattern in one dimension with an aperture width of 2x10^{-5}m, and screen distance of 1.2x10^{-4}m.

Figs.1-7 show the relative intensity against screen coordinate plot for Fresnel diffraction with different values of z. It is clear to see that as the screen gets closer to the aperture, the curve gets steeper, and starting from *4* near field patterns are observable. Because of the cyclic nature of the Fresnel equation, with decreasing z the run time of the code increased significantly. All the figures in part A shows a periodic diffraction pattern that was wider when z was greater and narrower when z smaller. This periodic pattern almost looks like a wave going up and down, breaking/splitting and merging back together. All figures including the near field diffraction ones had maxima at $x = 0m$ with negative gradients to either side of the x-axis as $|x|$ increased until other maxima were formed.

Part B:

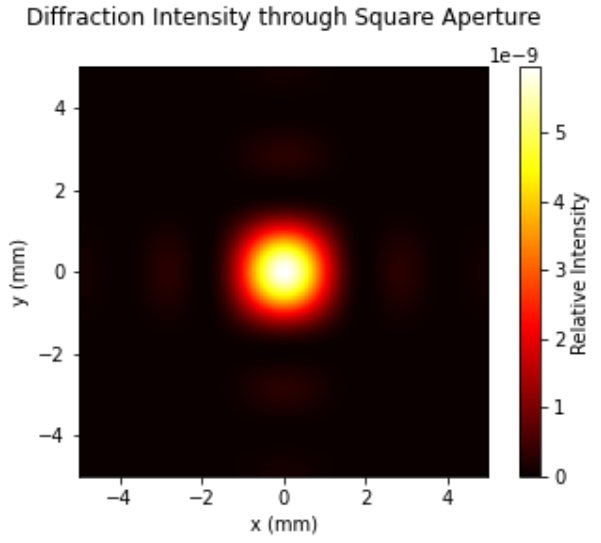


FIG. 8. Fresnel diffraction pattern through a square aperture with a width of 1x10^{-6}m, and screen distance of $0.002m$.

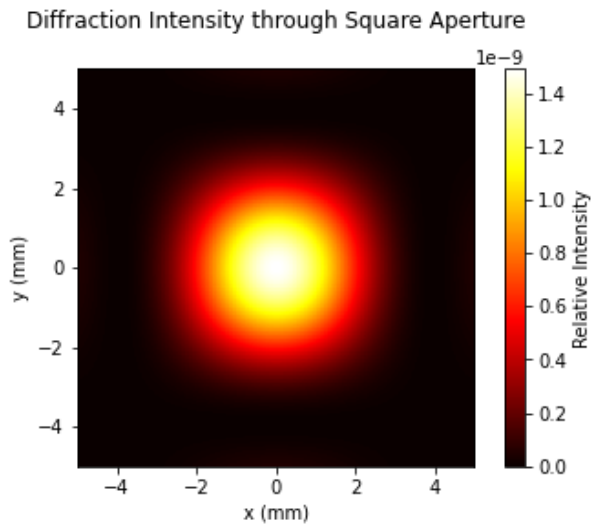


FIG. 9. Fresnel diffraction pattern through a square aperture with a width of $1\text{x}10^{-6}m$, and screen distance of $0.006m$.

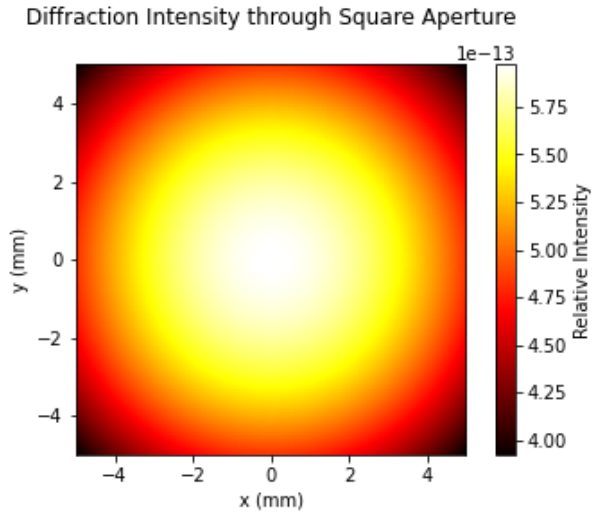


FIG. 10. Fresnel diffraction pattern through a square aperture with a width of $1\text{x}10^{-7}m$, and screen distance of $0.002m$.

Figs. 8-10 show the diffraction intensity of an incident beam through a square aperture. *Fig. 8* shows the best result obtained by the model for square aperture and it makes sense, since the wavelength of the incident light beam was also $1\text{x}10^{-6}m$. When the aperture width start to differ from the incident beam's wavelength, the diffraction pattern spreads which makes it harder to see, this can be seen in *fig.10*.
Fig. 9 makes sense intuitively, since the aperture width is still equal to the wavelength but the screen is further away, which causes the diffraction to spread. The patterns in the middle formed a circular shape as the incident light diffracted and their corners softened.

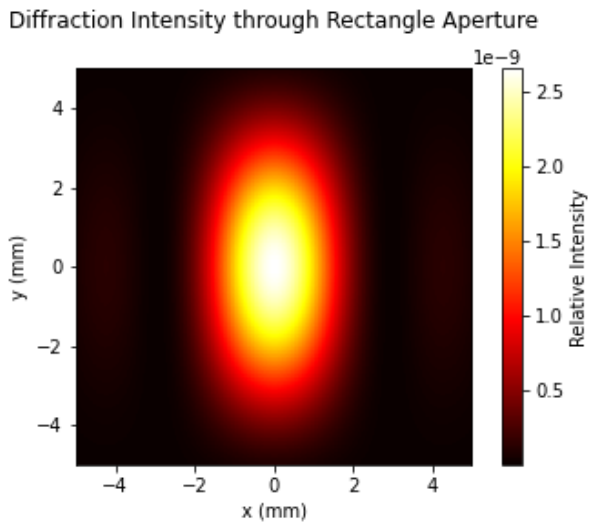


FIG. 12. Fresnel diffraction pattern through a rectangle aperture with a width of $1\text{x}10^{-6}m$, and screen distance of $0.006m$.

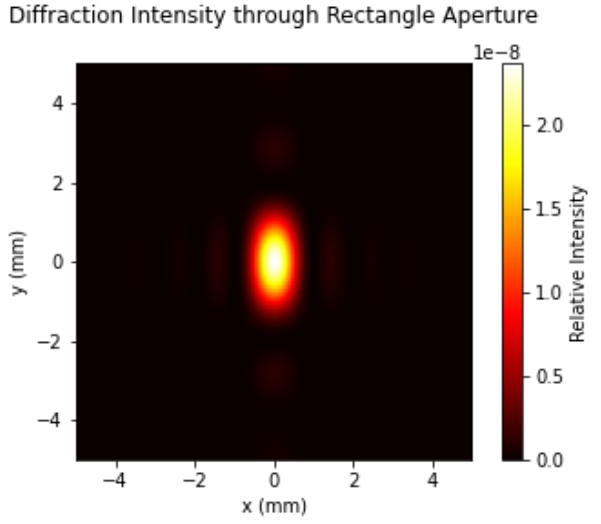


FIG. 11. Fresnel diffraction pattern through a rectangle aperture with a width of $1\text{x}10^{-6}m$, and screen distance of $0.003m$.

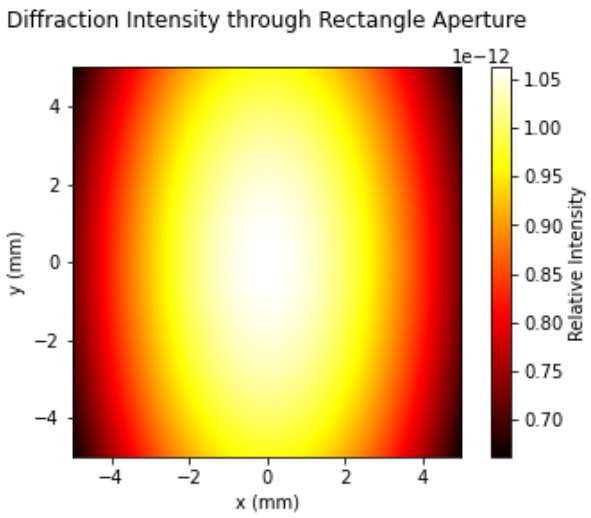


FIG. 13. Fresnel diffraction pattern through a rectangle aperture with a width of $1\text{x}10^{-7}m$, and screen distance of $0.003m$.

The rectangular aperture showed a similar diffraction pattern to the square but the constructive interference next to its longer sides are intuitively longer. Due to the its shape, the pattern in the middle formed an elliptic shape as the light started to diffract more.

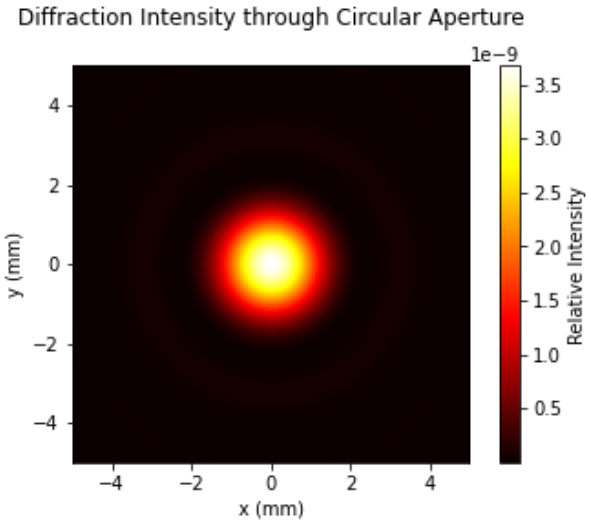


FIG. 14. Fresnel diffraction pattern through a circle aperture with a radius of $5\text{x}10^{-7}m$, and screen distance of $0.002m$.

Circle aperture with a ring of diffraction around the centre.

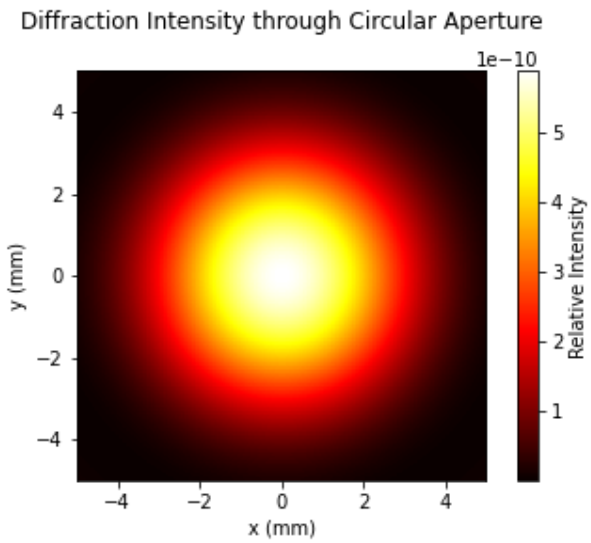


FIG. 15. Fresnel diffraction pattern through a circle aperture with a radius of $5\text{x}10^{-7}m$, and screen distance of $0.005m$.

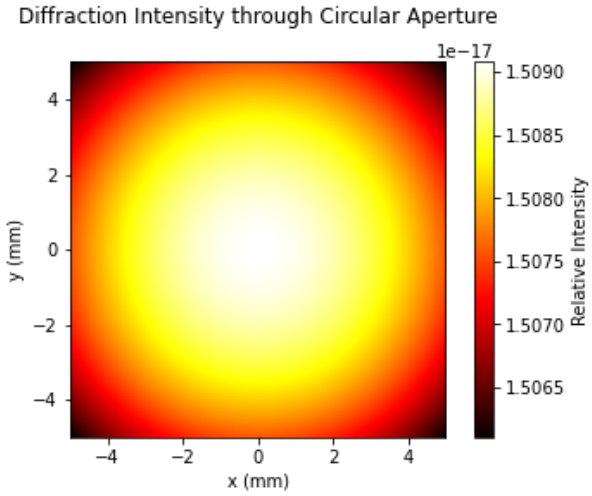


FIG. 16. Fresnel diffraction pattern through a circle aperture with a radius of $1\text{x}10^{-8}m$, and screen distance of $0.002m$.

The different shapes of apertures all followed the same rules when it comes to the various values of aperture width and screen distance were used. As the aperture width differed from the value for the wavelength and the screen got further away from the aperture the diffraction patterns got wider.

DISCUSSION

All figures included in parts A & B, showed a decrease in intensity as the screen moved away form the aperture, and got wider as the screen got further away from the aperture. Although it was simple and one dimensional, part A gave quite interesting periodic results with the varying variables. The diffraction through the two dimensional apertures were really hard to see, but their diffraction patterns were still noticeable. More data points could definitely be used in order to improve results in future models Despite looking simple, Part A's run time was already quite slow at 500 data points. In part B, 200 data points could be used, the model was able to produce images and it wasn't as slow as expected but for practicality reasons the numpoints for part B was kept at 100.
The model could definitely be optimised and improved, and other method could be implemented as well in order to get more accurate results for the diffraction patterns.
At times the code was giving errors when going over default the integration limit of SciPy, when this was tried to overcome the error tolerance of the code had to be increased but for the variables used in this report there was no critical difference between the methods of integration.

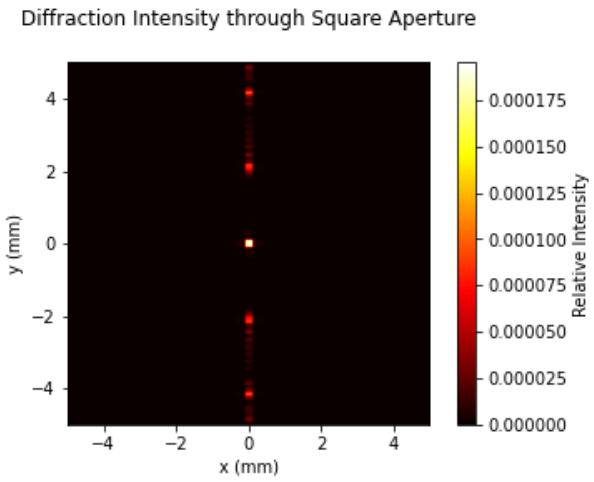


FIG. 17. Fresnel diffraction pattern through a square aperture with a width of $2\text{x}10^{-5}m$, and screen distance of $0.005m$.

APPENDIX

Part A, full equation used:

$$\begin{aligned} E(x,z) &= \frac{kE_0}{2\pi z}\int_{x_1'}^{x_2'} \exp\left[i\frac{k}{2z}(x-x')^2\right]dx' \\ &= \frac{kE_0}{2\pi z}\left[\int_{x_1'}^{x_2'} \cos\left(\frac{k}{2z}(x-x')^2\right)dx' \right. \\ &\quad \left. +i\int_{x_1'}^{x_2'} \sin\left(\frac{k}{2z}(x-x')^2\right]dx'\right) \end{aligned} \tag{4}$$

Name	Value
wavelength, λ	1×10^{-6} m
speed of light, c	$299,792,458$ m/s
permittivity of free space, ε_0	$8.85418782 \times 10^{-12}$ m^{-3} kg^{-1} s^4 A^2
incident electric field, E_0	3 V/m

TABLE I. Table of variables/constants used in the model.

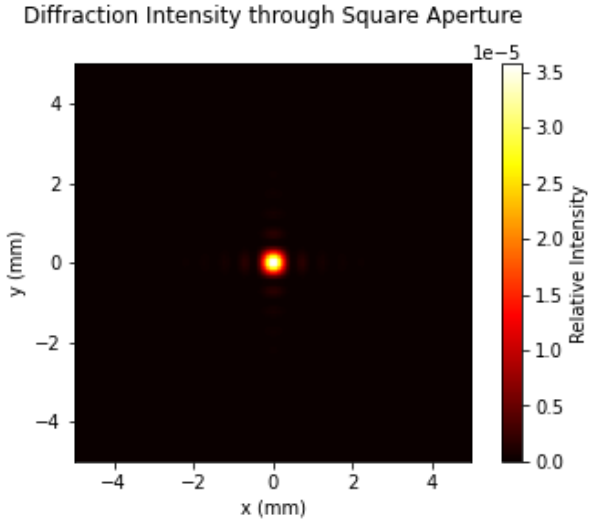


FIG. 18. Fresnel diffraction pattern through a square aperture with a width of $2\text{x}10^{-5}m$, and screen distance of $0.01m$.

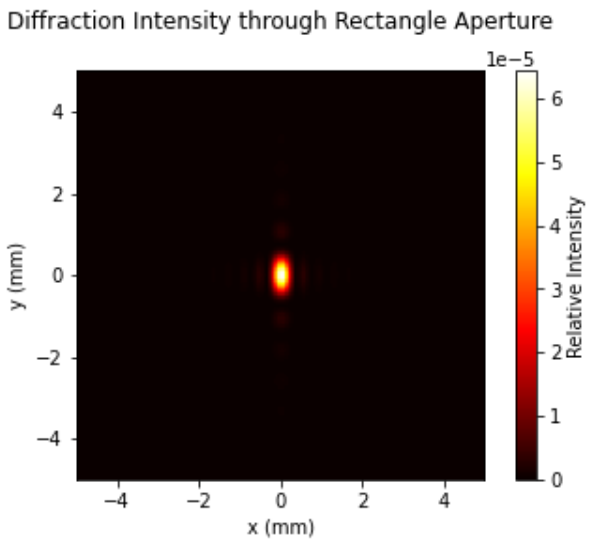


FIG. 19. Fresnel diffraction pattern through a rectangle aperture with a width of 2x$10^{-5}m$, and screen distance of $0.015m$.

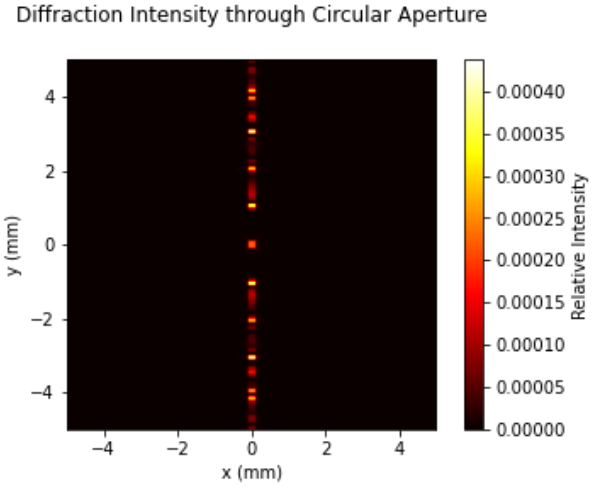


FIG. 20. Fresnel diffraction pattern through a rectangle aperture with a width of 2x$10^{-5}m$, and screen distance of $0.003m$.

REFERENCES

[1] Fowles, G.R. (2012). Introduction to Modern Optics. Newburyport: Dover Publications.
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[3] Francis Arthur Jenkins and Harvey Elliott White (1976). Fundamentals of Optics. McGraw-Hill Companies.